

Awesome LuaTeX

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1 Introduction

Welcome to Awesome LuaTeX! This document showcases a collection of useful LuaTeX functions and packages that enhance your LaTeX documents with dynamic content and powerful features.

LuaTeX combines the typesetting power of TeX with the flexibility of the Lua programming language, enabling you to create interactive and data-driven documents.

2 Packages

This section contains various LuaTeX packages and utilities for enhancing your documents.

2.1 GitHub

This subsection contains utilities for integrating GitHub data into your LaTeX documents.

2.1.1 GitHub Stars Fetcher

This package allows you to automatically fetch and display GitHub repository star counts in real-time.

Introduction The GitHub Stars Fetcher demonstrates how to use LuaTeX to fetch GitHub repository star counts dynamically during document compilation.

Load Lua Script First, load the Lua script in your document:

```
\directlua{dofile("packages/github/github-stars.lua")}
```

How It Works This script:

1. Uses shields.io API to fetch GitHub star count (bypasses GitHub API rate limit)
2. Parses SVG response to extract star count
3. Formats the output:
 - ≥ 1000 stars: displays as "x.xk+"
 - < 1000 stars: rounds down to nearest 10, displays as "xx+"

Usage Instructions Use the `\getgithubstars{owner/repo}` command in your LaTeX document.

Example:

```
\getgithubstars{torvalds/linux}
```

Compilation command:

```
lualatex --shell-escape your-document.tex
```

Note: Compilation requires the `--shell-escape` option to execute external shell commands.

Usage Examples

- **Linux Kernel:** 206k+ stars
- **Matchengine Rust:** 22 stars

2.1.2 GitHub Forks Counter

Get the number of repository forks.

Introduction The GitHub Forks Counter fetches the fork count for any public GitHub repository.

Load Lua Script Load the script in your document:

```
\directlua{dofile("packages/github/github-forks.lua")}
```

How It Works This script uses shields.io API to fetch fork counts and displays them in a readable format (e.g., "50k").

Usage Instructions Use the `\getgithubforks{owner/repo}` command:

```
\getgithubforks{facebook/react}
```

Usage Examples

- **React:** 50k+ forks
- **Vue:** 34k+ forks

2.1.3 GitHub Watchers Counter

Get the number of repository watchers.

Introduction The GitHub Watchers Counter fetches the watcher count for any public GitHub repository.

Load Lua Script

```
\directlua{dofile("packages/github/github-watchers.lua")}
```

How It Works Fetches watcher data from shields.io API and formats it appropriately.

Usage Instructions

```
\getgithubwatchers{owner/repo}
```

Usage Examples

- **React:** 6.7k+ watchers

2.1.4 GitHub Contributors Counter

Get the number of repository contributors.

Introduction The GitHub Contributors Counter shows how many people have contributed to a repository.

Load Lua Script

```
\directlua{dofile("packages/github/github-contributors.lua")}
```

How It Works Fetches contributor count from shields.io API.

Usage Instructions

```
\getgithubcontributors{owner/repo}
```

Usage Examples

- **React:** 411 contributors

2.1.5 GitHub Issues Counter

Get the number of open issues.

Introduction The GitHub Issues Counter displays the current number of open issues in a repository.

Load Lua Script

```
\directlua{dofile("packages/github/github-issues.lua")}
```

How It Works Fetches open issue count from shields.io API and extracts the number from "XXX open" format.

Usage Instructions

```
\getgithubissues{owner/repo}
```

Usage Examples

- **React:** 824 open issues

2.1.6 GitHub License Detector

Get the repository license type.

Introduction The GitHub License Detector identifies the license under which a repository is distributed.

Load Lua Script

```
\directlua{dofile("packages/github/github-license.lua")}
```

How It Works Fetches license information from shields.io API and displays the license name (e.g., "MIT", "Apache-2.0").

Usage Instructions

```
\getgithublicense{owner/repo}
```

Usage Examples

- **React:** MIT
- **Linux:** not identifiable by github

2.1.7 GitHub Release Version

Get the latest release version.

Introduction The GitHub Release Version fetcher displays the most recent release tag.

Load Lua Script

```
\directlua{dofile("packages/github/github-release.lua")}
```

How It Works Fetches the latest release version from shields.io API (e.g., "v19.2.0").

Usage Instructions

```
\getgithubrelease{owner/repo}
```

Usage Examples

- **React:** v19.2.0

2.1.8 GitHub Tag Version

Get the latest tag.

Introduction The GitHub Tag Version fetcher displays the most recent tag.

Load Lua Script

```
\directlua{dofile("packages/github/github-tag.lua")}
```

How It Works Fetches the latest tag from shields.io API.

Usage Instructions

```
\getgithubtag{owner/repo}
```

Usage Examples

- **React:** v19.2.0

2.1.9 GitHub Language Detector

Get the primary programming language.

Introduction The GitHub Language Detector identifies the main programming language used in a repository.

Load Lua Script

```
\directlua{dofile("packages/github/github-language.lua")}
```

How It Works Fetches language data from shields.io API and automatically capitalizes common language names (JavaScript, TypeScript, Python, etc.).

Usage Instructions

```
\getgithublanguage{owner/repo}
```

Usage Examples

- **React:** JavaScript
- **Linux:** C

2.1.10 GitHub Creation Date

Get the repository creation date.

Introduction The GitHub Creation Date fetcher displays when a repository was created.

Load Lua Script

```
\directlua{dofile("packages/github/github-created.lua")}
```

How It Works Fetches creation date from GitHub API and formats it as YYYY-MM-DD.

Usage Instructions

```
\getgithubcreated{owner/repo}
```

Usage Examples

- **React:** Created on 2013-05-24

2.1.11 Complete Example

Here's a comprehensive overview using all GitHub metrics:

- **Repository:** facebook/react
- **Stars:** 240k+
- **Forks:** 50k+
- **Watchers:** 6.7k+
- **Contributors:** 411
- **Open Issues:** 824
- **License:** MIT

- **Language:** JavaScript
- **Latest Release:** v19.2.0
- **Latest Tag:** v19.2.0
- **Created:** 2013-05-24